

**PERUNTHALAIVAR KAMARAJAR
INSTITUTE OF ENGINEERING AND TECHNOLOGY (PKIET)
NEDUNGADU (POST), KARAİKAL-609 603.
DEPARTMENT OF INFORMATION TECHNOLOGY
B.Tech (IT) – VI Semester - SOFTWARE ENGINEERING**

CAT - I

Time: 2 Hrs

Max: 50 Marks

PART-A - Answer ALL the questions (5 x 2=10)

1. What is meant by Software Engineering?
2. Define SDLC.
3. How Effort of Software development is measured?
4. What is the Goal of Requirement Analysis?
5. What is the main aim of Design Process?

PART-B - Answer ALL the questions (5 x 6=30)

6. Distinguish between program and professionally developed software.
7. Explain Prototype model for software development.
8. Discuss about the shortcomings of LOC metric.
9. Explain about organization and team structures.
10. Explain the characteristics of good SRS document.

PART-C - Answer ANY One questions (1 x 10=10)

11. With appropriate example, explain COCOMO model for software cost estimation.
Or
12. Enumerate different types of cohesion that a module in a design might exhibit.

DEPARTMENT OF INFORMATION TECHNOLOGY
CAT 1 - AUTOMATA AND FORMAL LANGUAGES

PART A- 2 X 5 = 10 Marks

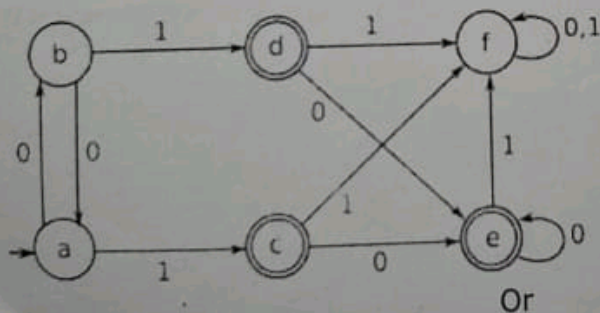
1. Write the Applications of Finite Automata.
2. State the difference between DFA and NFA.
3. Construct a Minimal DFA that accepts Epsilon.
4. Differentiate between Moore and Mealy Machine.
5. Write the tuple of 2DFA.

PART B- 5 X 6 = 10 Marks

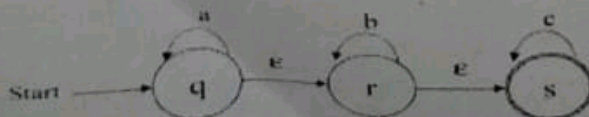
6. Design FA which accepts odd number of 1's and even number of 0's
7. Design FA to check whether given binary number is divisible by three.
8. Construct a Moore Machine that prints '1' which encounters the sequence of 'aab'.
9. Construct a Mealy Machine that prints 'a' which encounters either '11' or '00'.
10. With suitable example explain 2DFA.

PART C= 1 X 10 = 10 Marks

11. Minimize the following Automata.



12. Convert the following NFA with ϵ moves to DFA without ϵ moves.



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Dept of Information Technology

I Cycle test / III yr / WEB TECHNOLOGY

PART – A Answer All the Questions (4 x 2 = 8 Marks)

1. Define Web Browser.
2. State the advantages of HTML CSS.
3. Compare JavaScript with Java .
4. Define SMTP.

PART – B Answer All the Questions (4 x 8 = 32 marks)

1. Explain HTTP with example.
2. Discuss Web Server and its types with Architecture.
3. Discuss POP3 and IMAP and its advantages.
4. With an example explain JavaScript Functions.

PART – C Answer All the Questions (1 x 10 = 10 marks)

1. Compare SMTP POP3 IMAP and HTTP.

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Department of Information Technology
First Cycle Test
EPA101 Entrepreneurship
Section – A (5 x 4= 20 Marks)

1. Why the problem interview is essential? Justify it.
2. Mention the key components in value proposition canvas.
3. What are early adapters? State the characteristics and role of it,
4. Discuss the techniques of problem-solution test briefly.
5. Write the structure of MVP presentation.

Section – B (5 x 6 = 30 marks)

6. Explain the concept, characteristics, importance and types of business opportunity in detail.
7. Describe the stage process, key aspects of design thinking. Discuss the strategic importance of segmentation and targeting briefly.
8. Elucidate the key concept and benefits of blue ocean strategy with suitable example.
9. Illustrate the key components, principle and core benefits of Build-Measure – Learn feedback loop briefly. Discuss the concept and types of competition in entrepreneurship briefly.
10. Explain the various entrepreneurial styles and its benefits briefly. Discuss the types of market briefly.

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A Constituent College of PUDUCHERRY TECHNOLOGICAL UNIVERSITY (PTU),
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Time: 2 Hrs

CLOUD COMPUTING

Max: 50 Marks

PART-A

Answer ALL the questions

5*2=10

1. Define Cloud Computing.
2. List any four characteristics of cloud computing.
3. What is virtualization?
4. What is the difference between public and private cloud?
5. What is the role of open standards in cloud computing?

PART-B

Answer ALL the questions

5*6=30

6. Discuss the layers and types of cloud computing with diagram.
7. Explain Infrastructure as a Service (IaaS) with examples.
8. Explain Platform as a Service (PaaS) with examples.
9. Illustrate the concept of cloud computing stack with neat diagram.
10. Discuss about Software as a Service (SaaS) with examples.

PART-C

Answer ANY One question

1*10=10

11. Explain Cloud Infrastructure Management with neat diagram.

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I Cycle test / III yr / C# AND .NET PROGRAMMING

PART – A Answer All the Questions (4 x 2 = 8 Marks)

1. Define C#.Net.
2. Define Jagged Array and its advantages.
3. Compare Indexer and Properties.
4. Define Boxing and Unboxing.

PART – B Answer All the Questions (4 x 8 = 32 marks)

1. Write a C# program for sum of n numbers using function.
2. Discuss C# Structure with an example.
3. Discuss Indexer and properties with suitable example.
4. With an example explain Jagged Array .

PART – C Answer All the Questions (1 x 10 = 10 marks)

1. With an example Explain C# Delegates and Events.

1. What are the types of microwave ovens?
2. What are the advantages of microwave cooking?
3. What are the components of an air conditioning system?
4. How humidity and air cleanliness are controlled in an air conditioning system?

5. Draw the block diagram of microwave oven and briefly explain each block.

OR

6. Explain the working of a single chip microcontroller.
7. With the help of suitable diagram explain the working of LCD timer with alarm.

OR

8. Explain the working of multicavity magnetism.
9. Distinguish between all-air and all water air conditioning system with suitable diagram.

OR

10. Write short notes on: (a) Combination systems (b) Split air conditioners
(c) Refrigerants (d) Evaporative condenser
11. Explain the working of domestic refrigerator.

OR

12. Briefly explain how the standards of temperature, humidity, cleanliness and motion is maintained in an air conditioner and also explain the components of air conditioning system briefly.

13. Explain the following:

(a) Wave reflector system (b) Dual emission system (c) Fail safe system (d) 3D microwaves

OR

14. Explain in detail the refrigeration cycle.